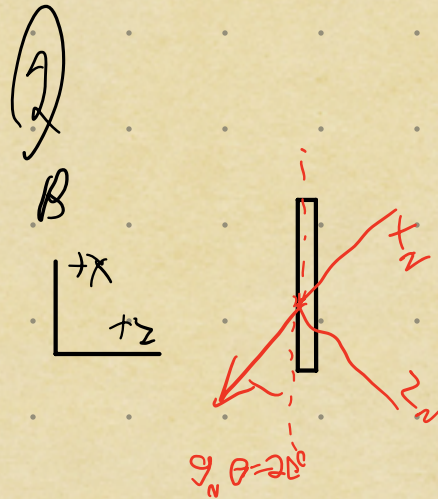
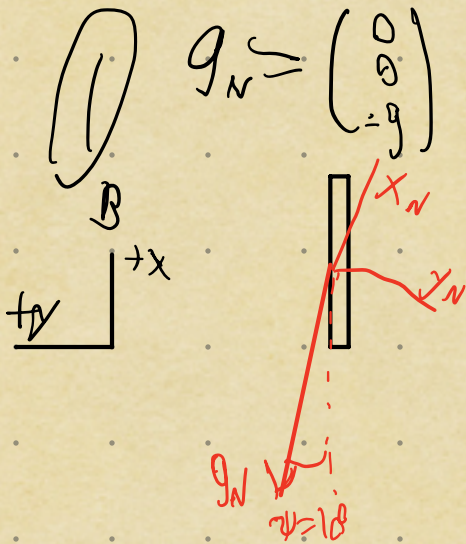


Given: $\theta = 20^\circ$, $\psi = 10^\circ$, express g_N in body frame

2, Y, X order



① $g_{N1} = \begin{bmatrix} -g \cos(10^\circ) \\ g \sin(10^\circ) \\ 0 \end{bmatrix} = R_2 g_N$

② $g_B = \begin{bmatrix} -g \cos(20^\circ) \cos(10^\circ) \\ g \sin(10^\circ) \\ -g \sin(20^\circ) \end{bmatrix} = R_Y g_{N1} = R_Y R_2 g_N$

$g = 9.81 \text{ m/s}^2$

$g_B = \begin{bmatrix} -9.08 \\ 1.70 \\ -3.36 \end{bmatrix} \text{ m/s}^2$

Pretty sure answer provided on website is wrong

because provided answer has y-component as 0; This is clearly false.

Documentation: I used MATLAB to perform the sine and cosine calculations.

Once I completed `euler2dcm.m`, I used it to check my work and saw that I did not define my yaw coordinate system correctly, fixing my error.

